

TAFWELA

User Stories & Use Cases

(DPEI) | 2026

1. Purpose

This document defines User Stories and Use Cases for the TAFWELA application. User Stories describe the system from the end user's perspective, while Use Cases provide detailed interaction flows between actors and the system.

2. System Actors

#	Actor	Type	Description
1	Driver	Primary Actor	Egyptian driver using the app to find fuel stations
2	Station Staff	Primary Actor	Fuel station employee who updates congestion status
3	Firebase System	Secondary Actor	Backend system handling auth, data storage, and real-time sync
4	Google Maps	External Actor	Provides navigation and map rendering services

3. User Stories

3.1 Driver User Stories

ID	User Story	Priority	Feature
US-01	As a driver, I want to see nearby fuel stations on a map so that I can identify options close to my location.	Must Have	Map View
US-02	As a driver, I want to see color-coded markers (Green/Yellow/Red/Gray) so that I can instantly know each station's congestion level.	Must Have	Map Markers
US-03	As a driver, I want the app to automatically detect my GPS location so that I don't have to enter my address manually.	Must Have	GPS
US-04	As a driver, I want to tap a station marker to view its details (name, address, distance, status, hours) so that I can make an informed decision.	Must Have	Station Details
US-05	As a driver, I want to tap a Navigate button so that Google Maps opens with turn-by-turn directions to the station.	Must Have	Navigation
US-06	As a driver, I want to search for a station by name or area so that I can find a specific station quickly.	Should Have	Search
US-07	As a driver, I want to filter stations by congestion status so that I only see stations with low congestion.	Should Have	Filter
US-08	As a driver, I want to view stations in a list view as an alternative to the map so that I can compare options easily.	Should Have	List View
US-09	As a driver, I want to save favourite stations so that I can quickly access the ones I visit most often.	Should Have	Favourites
US-10	As a driver, I want to register and log in with my email so that my favourites are saved securely.	Must Have	Auth

US-11	As a driver, I want to share station details with others so that I can help friends or family find a good station.	Could Have	Share
US-12	As a driver, I want to see an onboarding screen when I first open the app so that I understand how to use it.	Should Have	Onboarding

3.2 Station Staff User Stories

ID	User Story	Priority	Feature
US-13	As a station staff member, I want to log in with a dedicated account so that I can access the Staff Panel securely.	Must Have	Staff Auth
US-14	As a station staff member, I want to update the congestion status of my station (Low/Medium/High/No Fuel) so that drivers can see real-time information.	Must Have	Status Update
US-15	As a station staff member, I want to see a confirmation dialog before updating the status so that I don't make accidental changes.	Should Have	Confirmation
US-16	As a station staff member, I want to see the last update time and history log so that I can track when changes were made.	Should Have	History Log
US-17	As a station staff member, I want to see my station name and profile info on the dashboard so that I know I am logged into the correct station.	Must Have	Staff Profile

4. Use Cases

UC-01: View Nearby Stations on Map

Use Case ID	UC-01
Use Case Name	View Nearby Stations on Map
Actor	Driver
Preconditions	Driver has opened the app and granted GPS permission.
Main Flow	1. App detects driver's GPS location automatically. 2. System queries Firestore for stations within the configured radius. 3. Map displays stations as color-coded markers (Green/Yellow/Red/Gray). 4. Driver sees their location pin on the map.
Alternative Flow	A1: GPS permission denied → App displays a prompt asking the driver to enable location manually. A2: No stations found within radius → App shows 'No stations found nearby' message.
Postconditions	Driver can see all nearby stations and their congestion levels on the map.
Related Stories	US-01, US-02, US-03

UC-02: Navigate to a Fuel Station

Use Case ID	UC-02
Use Case Name	Navigate to a Fuel Station
Actor	Driver
Preconditions	Driver has opened the map and can see at least one station marker.
Main Flow	1. Driver taps a station marker on the map. 2. System displays the Station Details screen (name, address, distance, status, hours). 3. Driver taps the Navigate button. 4. System launches Google Maps with the station's coordinates as the destination. 5. Google Maps provides turn-by-turn directions.
Alternative Flow	A1: Google Maps not installed → App opens Google Maps in the browser instead. A2: Station details failed to load → App shows an error message and retry option.
Postconditions	Driver is navigating to the selected fuel station via Google Maps.
Related Stories	US-04, US-05

UC-03: Update Station Congestion Status (Staff)

Use Case ID	UC-03
Use Case Name	Update Station Congestion Status
Actor	Station Staff
Preconditions	Staff member is logged in to the Staff Panel with an authorized account.
Main Flow	1. Staff member opens the Staff Dashboard. 2. System displays current station name, congestion status, and last update time. 3. Staff member selects a new congestion level (Low / Medium / High / No Fuel). 4. System displays a confirmation dialog: 'Are you sure you want to update the status?' 5. Staff member confirms. 6. System updates Firestore and records the change with timestamp and staff ID. 7. All drivers viewing the map see the updated marker color in real time (< 1 second).
Alternative Flow	A1: Staff member cancels the confirmation → No changes are made. A2: Firestore write fails → App shows error message and prompts retry.
Postconditions	Station congestion status is updated in Firestore and reflected on all driver maps in real time.
Related Stories	US-13, US-14, US-15, US-16

UC-04: Search and Filter Stations

Use Case ID	UC-04
Use Case Name	Search and Filter Stations
Actor	Driver
Preconditions	Driver is on the Map or List View screen.
Main Flow	1. Driver taps the Search bar and enters a station name or area. 2. System filters the displayed stations in real time as the driver types. 3. Driver optionally applies a filter (e.g., Green only). 4. System updates the list/map to show only matching stations. 5. Driver selects a station from the results.
Alternative Flow	A1: No stations match the search query → App displays 'No results found'. A2: Driver clears the search → Full station list is restored.
Postconditions	Driver sees a filtered list of stations matching their search criteria.
Related Stories	US-06, US-07, US-08

UC-05: Register and Login

Use Case ID	UC-05
Use Case Name	Register and Login
Actor	Driver / Station Staff
Preconditions	User has opened the app for the first time or has been logged out.
Main Flow	1. User selects Register or Login on the welcome screen. 2. For Register: User enters email and password, system creates account via Firebase Auth. 3. For Login: User enters credentials, Firebase Auth verifies them. 4. System determines user role (Driver or Staff) and redirects to the appropriate screen. 5. Driver is taken to the Map screen. Staff is taken to the Staff Dashboard.
Alternative Flow	A1: Invalid credentials → App shows 'Incorrect email or password' error. A2: Email already registered → App shows 'Account already exists' and prompts login. A3: Weak password → App shows password strength requirements.
Postconditions	User is authenticated and directed to the correct screen based on their role.
Related Stories	US-10, US-13

UC-06: Save and View Favourite Stations

Use Case ID	UC-06
Use Case Name	Save and View Favourite Stations
Actor	Driver
Preconditions	Driver is logged in and viewing a station's details.
Main Flow	1. Driver taps the Favourite (heart) icon on a station's details screen. 2. System saves the station to local storage (shared_preferences). 3. Driver navigates to the Favourites screen. 4. System displays all saved favourite stations with their current congestion status.
Alternative Flow	A1: Driver taps the heart icon again → Station is removed from favourites. A2: Favourites list is empty → App displays 'No favourites saved yet'.
Postconditions	Station is saved to/removed from the driver's favourites list.
Related Stories	US-09

5. User Stories Summary

Actor	Must Have	Should Have	Could Have	Total
Driver	4	6	1	12
Station Staff	3	2	0	5
TOTAL	7	8	1	17